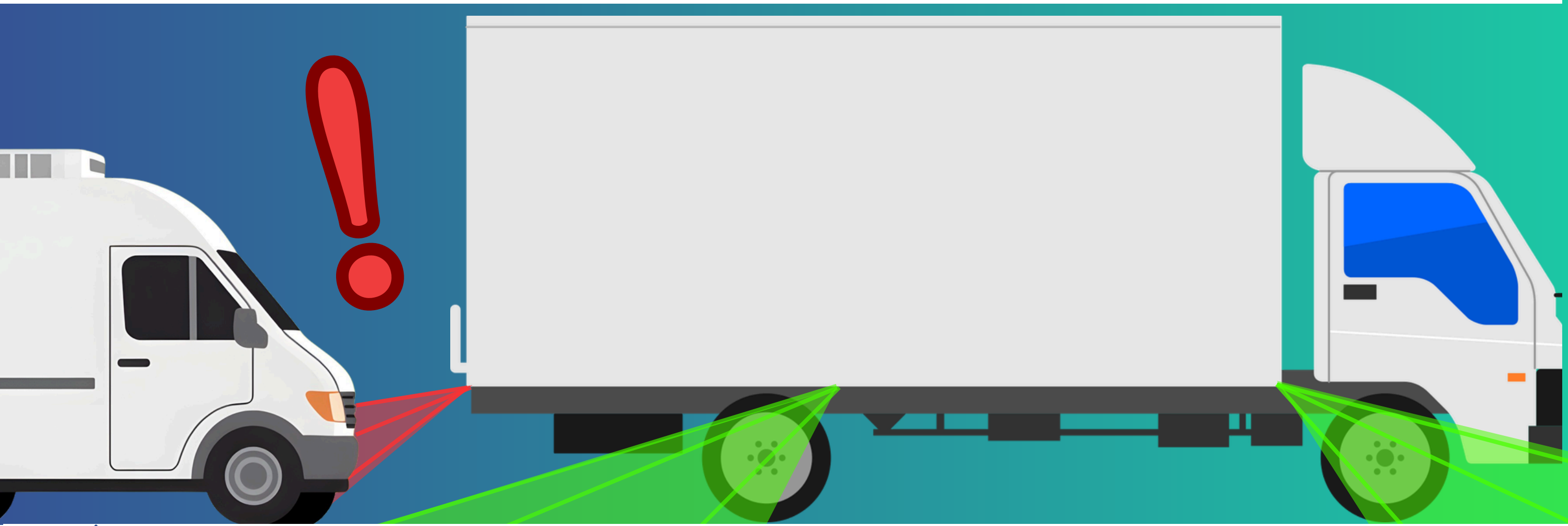


Telemetry to Transmission:

Override Detection and Classification Using CAN Signals

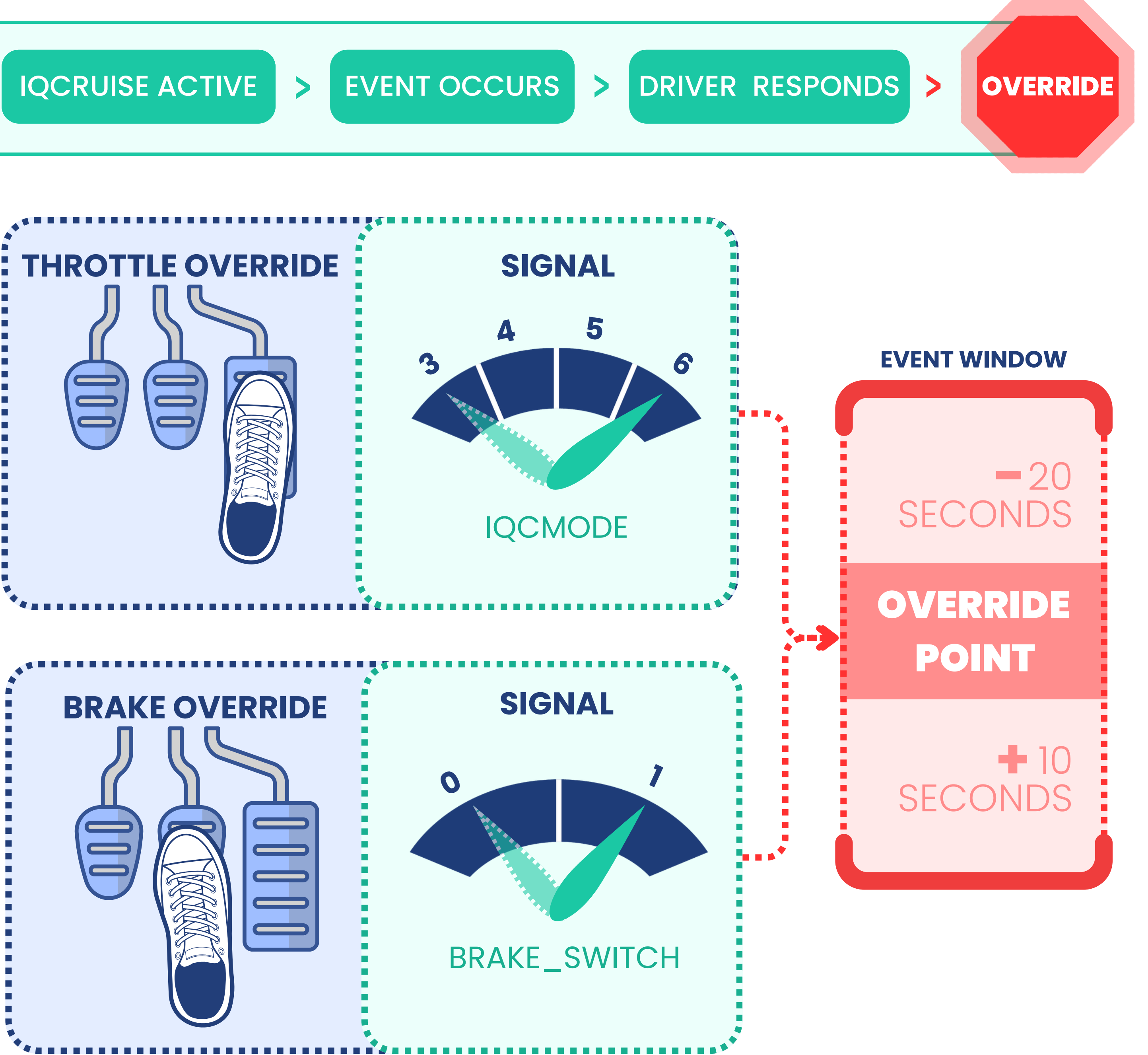


01 PROBLEM + OBJECTIVE

Traxen Inc. deploys an **Advanced Driver Assistance System (ADAS)** referred to as iQCruise on commercial truck fleets for fuel-efficient cruise control. Drivers frequently **override the system** via throttle or brake, but the raw sensor logs provide no explanation of **why**.

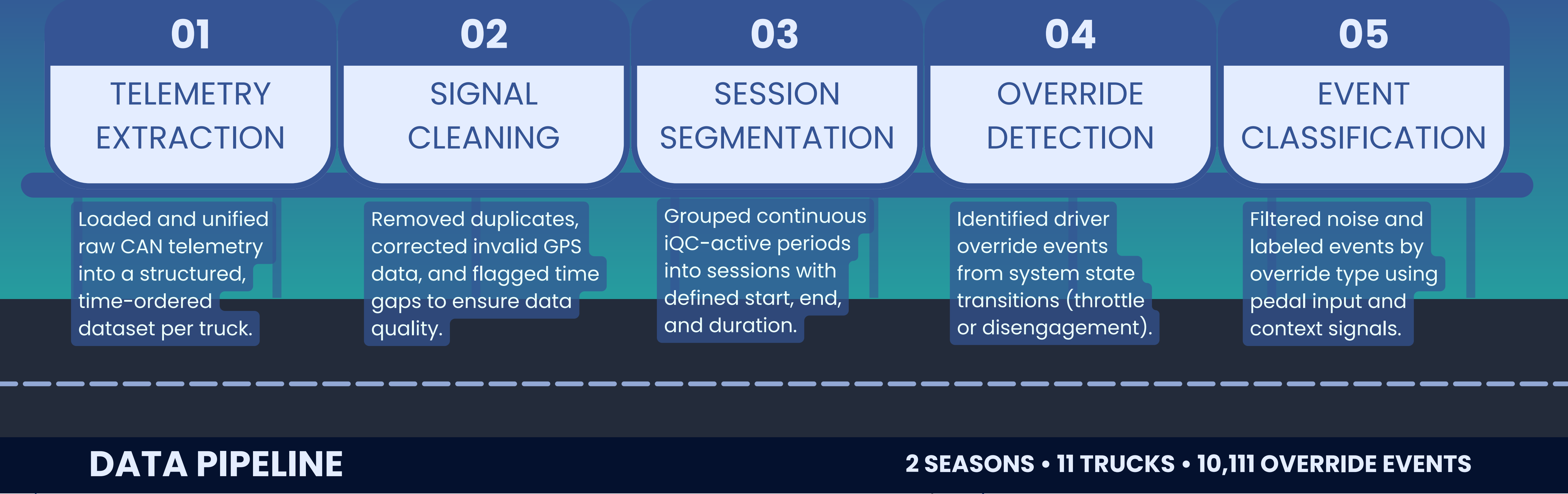
Our objective is to build a pipeline that extracts override events from time-series CAN (controller area network) bus data, engineers contextual features from pre-override signal windows, and applies unsupervised learning to **classify them to surface root causes**.

02 IQCRUISE + OVERRIDE DETECTION

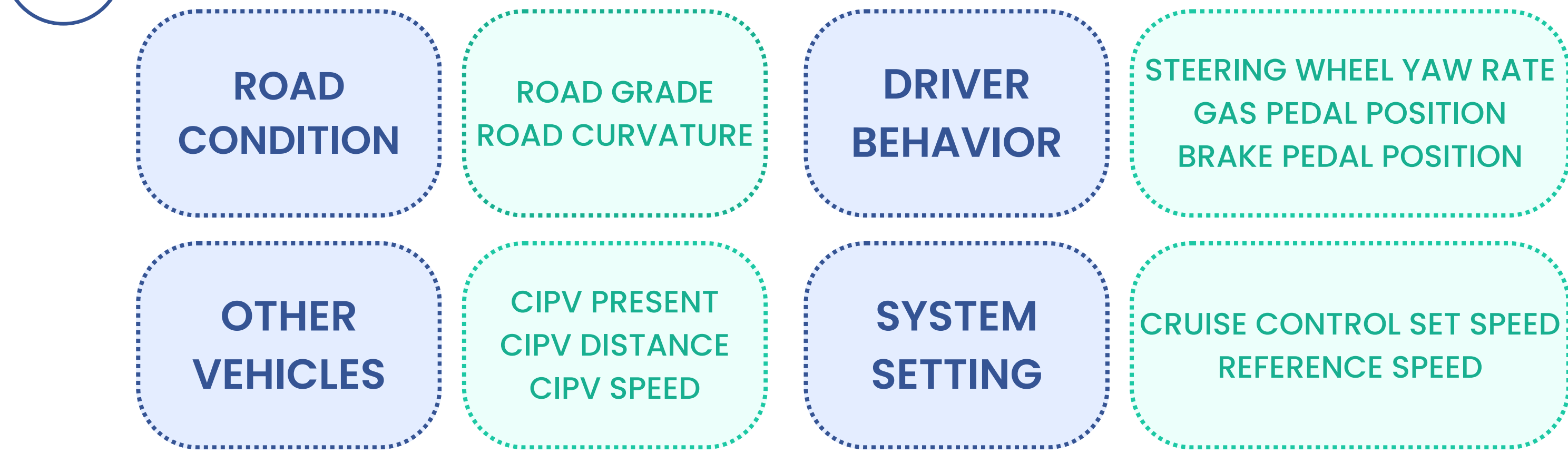


FUTURE WORK

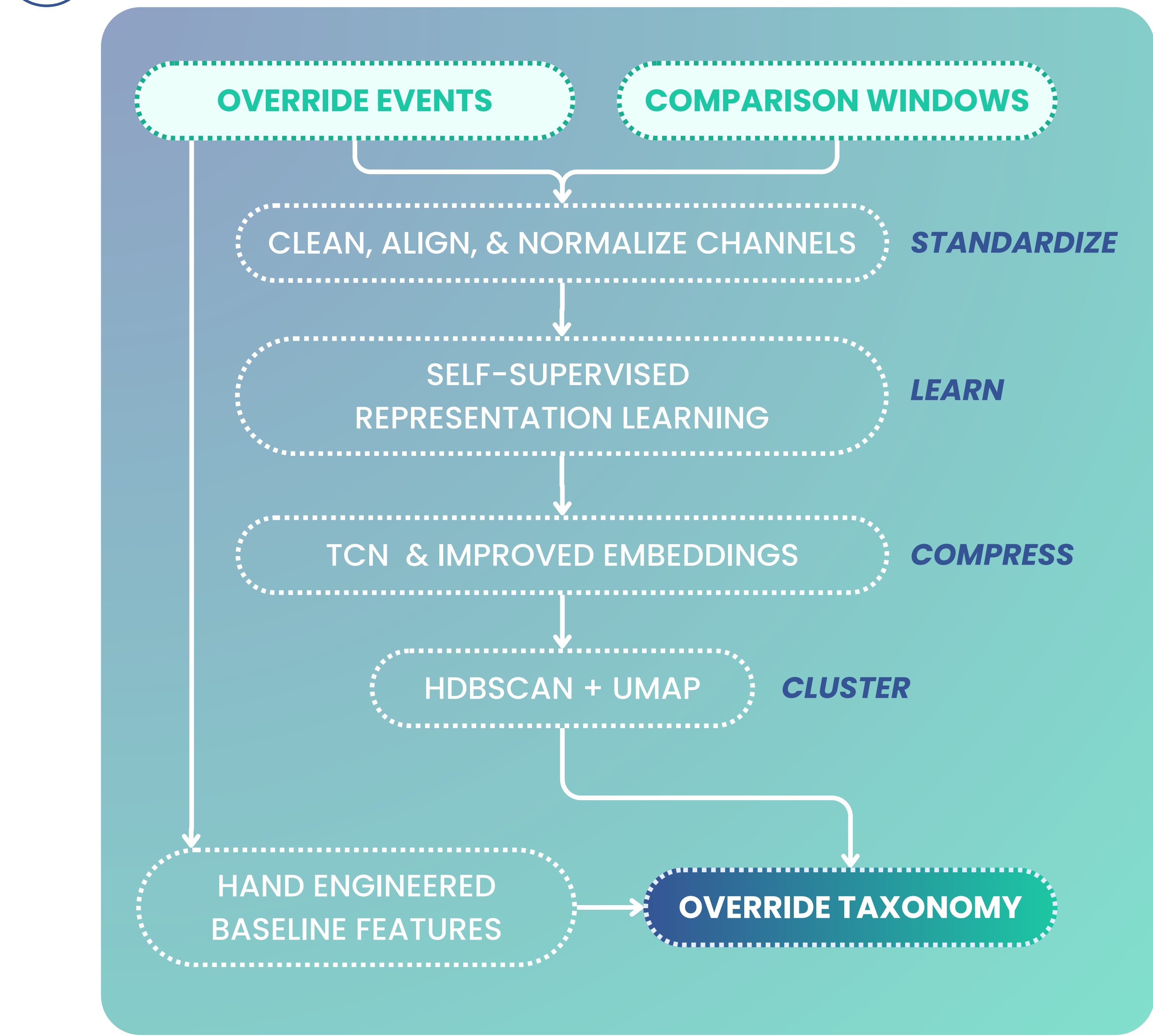
- Validate taxonomy with Traxen domain experts and clients.
- Deploy as real-time anomaly flagging on incoming iQCruise data streams.



03 SIGNAL EXTRACTION



04 CLASSIFICATION LOGIC



RELATED LITERATURE

Milardo, S., Rathore, P., Santi, P., Buteau, R., & Ratti, C. (2022). An unsupervised approach for driving behavior analysis of professional truck drivers. In A. L. Martins, J. C. Ferreira, & A. Kocian (Eds.), *Intelligent transport systems* (Vol. 426, pp. 47–60). Springer. https://doi.org/10.1007/978-3-030-97803-3_4

Rosenfeld, Avi & Bareket, Zvi & Goldman, Claudia & Kraus, Sarit & Leblanc, David & Tsimhani, Orner. (2014). Learning Driver's Behavior to Improve Adaptive Cruise Control. *Journal of Intelligent Transportation Systems*. 18. 10.1080/15472450.2014.889960. <https://doi.org/10.1080/15472450.2014.889960>

Sanna, C.. *Driver_behaviour_modelling_using_CAN_bus_data* [Computer software]. GitHub. https://github.com/chethansanna/driver_behaviour_modelling_using_CAN_bus_data



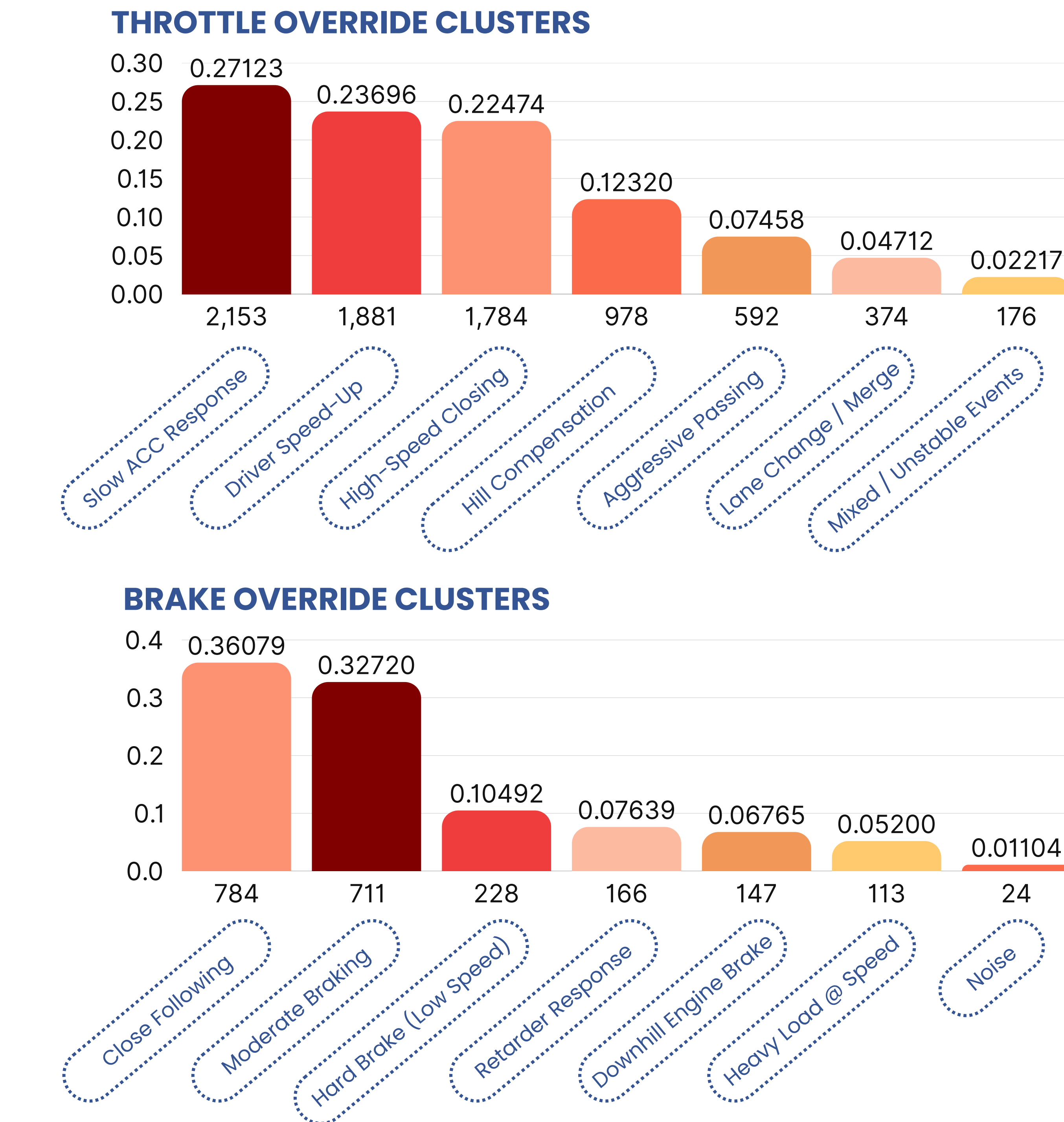
Authors
Jonathan McMillan
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Client
TRAXEN®

05 OUTPUTS & DIAGNOSTICS

Using UMAP dimensionality reduction and HDBSCAN clustering, we identified 13 root cause clusters across throttle and brake overrides, **without any preexisting human labels**.

Each cluster represents a distinct driving scenario that triggers **iQCruise disengagement**. Throttle overrides are dominated by system response lag and deliberate speed adjustments, while brake overrides reflect traffic-following behavior and road grade responses. **These clusters give fleet operators, for the first time, a data-driven answer to why drivers override** — enabling targeted system tuning, driver coaching, and route-level intervention.



Truck
Override
Animations
(Active):